

11.5. Let consider $\sum_{n=1}^{\infty} (-1)^n b_n$ is an alternating series then:
 If $\lim_{n \rightarrow \infty} b_n = 0$ and b_n is a decreasing sequence then
 $\sum_{n=1}^{\infty} (-1)^n b_n$.

Example: Let consider $\sum_{n=1}^{\infty} \frac{(-1)^n}{n}$. this is a convergent series b/c.
 $b_n = \frac{1}{n}$, $\lim_{n \rightarrow \infty} b_n = 0$ and $n+1 > n \Rightarrow \frac{1}{n} \geq \frac{1}{n+1}$
 So b_n is a decreasing sequence then A-T
 $\sum_{n=1}^{\infty} \frac{(-1)^n}{n} < \infty$.

Absolute Convergence and Conditional convergence.

Let consider series $\sum a_n$ and also $\sum |a_n|$

$$\sum_{n=1}^{\infty} |a_n| = |a_1| + |a_2| + |a_3| + \dots$$

Def. A series $\sum a_n$ is absolutely convergent if $\sum |a_n|$ is convergent.

Example: Let $\sum_{n=1}^{\infty} \frac{(-1)^n}{n^3}$ and $\sum_{n=1}^{\infty} \left| \frac{(-1)^n}{n^3} \right| = \sum_{n=1}^{\infty} \frac{1}{n^3}$ is convergent b/c it is a p-series $p=3 > 1$. So, $\sum_{n=1}^{\infty} \frac{(-1)^n}{n^3}$ is absolutely convergent.

Def: A series $\sum a_n$ is conditionally convergent if it is convergent but $\sum |a_n|$ is divergent.

Example: We know that $\sum_{n=1}^{\infty} \frac{(-1)^n}{n}$ is convergent but
 $\sum_{n=1}^{\infty} \left| \frac{(-1)^n}{n} \right| = \sum_{n=1}^{\infty} \frac{1}{n}$ is divergent
 then, $\sum_{n=1}^{\infty} \frac{(-1)^n}{n}$ is conditionally convergent.

Thm. If a series $\sum a_n$ is absolutely convergent, then it is convergent.

Example: Determine whether the series

$$\sum_{n=1}^{\infty} \frac{\cos n}{n^2}$$

is convergent or divergent.

Sln. $\sum_{n=1}^{\infty} \left| \frac{\cos n}{n^2} \right| = \sum_{n=1}^{\infty} \frac{|\cos n|}{n^2}$

Note that

$$\frac{|\cos n|}{n^2} \leq \frac{1}{n^2}$$

and $\sum_{n=1}^{\infty} \frac{1}{n^2}$ is convergent by p-series $p=2>1$. and using Direct comparison then $\sum_{n=1}^{\infty} \frac{|\cos n|}{n^2}$ is convergent.

So $\sum_{n=1}^{\infty} \frac{\cos n}{n^2}$ is absolutely convergent and by the thm it is convergent.

Problem 1. Determine whether the series is absolutely convergent, conditionally convergent, or divergent.

i). $\sum_{n=1}^{\infty} \frac{1 + 2 \sin n}{n^3}$

ii). $\sum_{n=1}^{\infty} \frac{\cos n\pi}{3n+2}$

Sln: ii). $\sum_{n=1}^{\infty} \left| \frac{\cos n\pi}{3n+2} \right| = \sum_{n=1}^{\infty} \frac{|\cos n\pi|}{3n+2} = \sum_{n=1}^{\infty} \frac{1}{3n+2}$ this is divergent!

$$\sum_{n=1}^{\infty} \frac{\cos n\pi}{3n+2} = \sum_{n=1}^{\infty} \frac{(-1)^n}{3n+2}$$

By A.T $\sum_{n=1}^{\infty} \frac{\cos n\pi}{3n+2}$ is

$$\left\{ \begin{array}{l} b_n = \frac{1}{3n+2} : \lim_{n \rightarrow \infty} b_n = 0 \checkmark \\ n+1 > n \Rightarrow 3(n+1) > 3n \\ \Rightarrow 3(n+1)+2 > 3n+2 \\ \Rightarrow \frac{1}{3n+2} > \frac{1}{3(n+1)+2} \end{array} \right.$$

convergent then $\sum_{n=1}^{\infty} \cos n$ is conditionally convergent. $b_n > b_{n+1}$.

i). Case 1. $|1 + 2 \sin n| \leq 3$ then

$$\frac{|1 + 2 \sin n|}{n^3} \leq \frac{3}{n^3}$$

and $\sum_{n=1}^{\infty} \frac{3}{n^3} = 3 \sum_{n=1}^{\infty} \frac{1}{n^3}$ is convergent b/c it is a p-series

with $p=3 > 1$.

So, by Direct comparison then $\sum_{n=1}^{\infty} \frac{|1 + 2 \sin n|}{n^3}$ is convergent.

Hence, $\sum_{n=1}^{\infty} \frac{1 + 2 \sin n}{n^3}$ is absolutely convergent and therefore

$$\sum_{n=1}^{\infty} \frac{1 + 2 \sin n}{n^3} \text{ is convergent.}$$

Case 2. $\sum_{n=1}^{\infty} \frac{1}{n^3} + \sum_{n=1}^{\infty} \frac{2 \sin n}{n^3} = \sum_{n=1}^{\infty} \frac{1 + 2 \sin n}{n^3}$ if $\sum_{n=1}^{\infty} \frac{1}{n^3}$ and

$\sum_{n=1}^{\infty} \frac{2 \sin n}{n^3}$ is convergent. Indeed, $\sum_{n=1}^{\infty} \frac{1}{n^3}$ is convergent by p-series

and $\sum_{n=1}^{\infty} \frac{2 \sin n}{n^3} \leq \sum_{n=1}^{\infty} \frac{2}{n^3}$ which is convergent as well

then $\sum_{n=1}^{\infty} \frac{1 + 2 \sin n}{n^3}$ is convergent. is it absolutely convergent?

$$\frac{|1 + 2 \sin n|}{n^3} \leq \frac{|1| + 2 |\sin n|}{n^3} = \frac{1}{n^3} + 2 \frac{|\sin n|}{n^3}$$

and since $\sum_{n=1}^{\infty} \frac{1}{n^3} < \infty$ and $\sum_{n=1}^{\infty} \frac{2 |\sin n|}{n^3}$ is convergent

then $\sum_{n=1}^{\infty} \left(\frac{1}{n^3} + 2 \frac{|\sin n|}{n^3} \right)$ is convergent by the comparison then $\sum_{n=1}^{\infty} \frac{|1 + 2 \sin n|}{n^3}$ is convergent.

Hence $\sum_{n=1}^{\infty} \frac{1+2 \sin n}{n^3}$ is absolutely convergent.